Slow sound: An undergraduate lab experience for critical thinking

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Slow sound: An undergraduate lab experience for critical thinking

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In a simple time of flight experiment, the speed of sound is observed to travel about 5% slower inside a corrugated irrigation pipe as compared with measurements conducted in open air. Although some theoretical understanding for this retardation is provided, the focus of this paper concerns how this finding might be incorporated into an undergraduate lab activity that challenges student understanding of error analysis and its significance. We propose an inquiry-based format, recently championed by others, wherein students are challenged to decide how to resolve their observation of a slower speed in a second round of investigations. © *2020 American Association of Physics Teachers.*
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I. INTRODUCTION

Recent research suggests that traditional introductory lab courses are failing to impart the intended goal of teaching students to think like a scientist.[1–5] Traditional lab activities that are guided and content-focused often tell students what to do and when to do it and in so doing leave little or no space for students to make decisions, plan procedures, or otherwise think independently. Even when these "cookbook"-style lab activities are employed with the sole intention of reinforcing course content from an accompanying lecture, they fail to achieve that simple goal as well.[1]

But change is on the way. A recent AAPT panel released recommendations[6] for the undergraduate physics laboratory curriculum, for both majors and non-majors, that emphasize six skills that practicing scientists rely on: Constructing knowledge, modeling, designing experiments, developing technical and practical lab skills, analyzing and visualizing data, and communicating physics. At the same time, a handful of groups are evaluating innovative options to the traditional lab activity that are designed to promote quantitative critical thinking skills for student participants. These "inquiry"-based labs challenge students to reason about their measurements (and uncertainty) and make decisions about how to proceed and what to believe.

A good example is the inquiry lab described in the paper by Holmes and Wieman,[1] in which students are tasked to measure the period of a simple pendulum at two different angles to determine if the period is the same or different. In their first attempt, student uncertainty is usually too large to detect any difference leading students to confirm their pre-existing belief (based on their textbook or lecture) that the period of a simple pendulum is independent of the angle. But then students are pressed to redesign the experiment to reduce their uncertainty and repeat the measurement. Students report being surprised to find that, when the uncertainty is sufficiently small, a small, but statistically significant difference appears in the pendulum's period.

While changes like these are helping to improve the introductory labs, it seems that far less has filtered up into those upper division lab courses taken by students majoring in physics. There, the AAPT recommendations remain applicable, but the learning goals extend to include more advanced statistics, modeling, and skills with common laboratory equipment (e.g., meters and oscilloscopes). For many years, the sophomore lab here at Creighton was designed merely to complement the course on modern physics. Students performed classic experiments from the earlier 1900s (e.g., electron charge-to-mass ratio or photoelectric effect) using commercially available equipment and guided by a handout. However, during a recent reorganization of the lab curriculum, this course was replaced by a two credit course (3 h lab plus 1 h lecture per week) titled "Research Methods." The classic experiments have been replaced by lab activities that are tailored to help students develop a set of specific experimental skills (e.g., error analysis, using multimeters and oscilloscopes, curve-fitting, programming with Arduinos and experiment design) that are in line with recent AAPT recommendations.[5,6] Much like the inquiry-based labs outlined above, at the heart of each activity is a measurement "target" which is conceptually straightforward in terms of the physics being considered, so that greater emphasis may be placed on learning fundamentals of data handling, error analysis, and the procedures of a trained scientist.

Here, we report on one of these activities, measuring the speed of sound by the time of flight technique. Although the activity was originally intended to help students learn how to operate an oscilloscope, our finding that sound is "slowed" when traveling inside a corrugated pipe has the potential to provide a deeper learning experience. In a way that is reminiscent of the inquiry-based pendulum lab discussed above, students are confronted by an unexpected result and must evaluate their error and apply quantitative critical thinking to reason out its validity.

II. METHOD

Measuring the speed of sound by time of flight provides an excellent first exposure to using an oscilloscope and is conceptually simple: Sound pulses travel some distance from a speaker to a microphone and the time delay is observed by comparing two signals on the oscilloscope. The experimental setup is outlined in Fig. 1. In our case, a square wave (typically 2–5 Hz) from a function generator is sent to one channel of an oscilloscope and also to a small speaker (0.1 W, 8 Ω, 5 cm diameter) located a few inches from one end of a 1.2 m length of inexpensive 10 cm diameter corrugated irrigation pipe.[7] A simple microphone (constructed from an electret element in combination with 1 μF capacitor and 1 kΩ resistor as illustrated in Fig. 1) is housed in a 4 × 6 × 8 cm enclosure and affixed to the end of a meterstick which can be slid into the corrugated pipe. The output signal of the microphone is directed by a coaxial cable to a second channel of the oscilloscope. Triggering is established on the

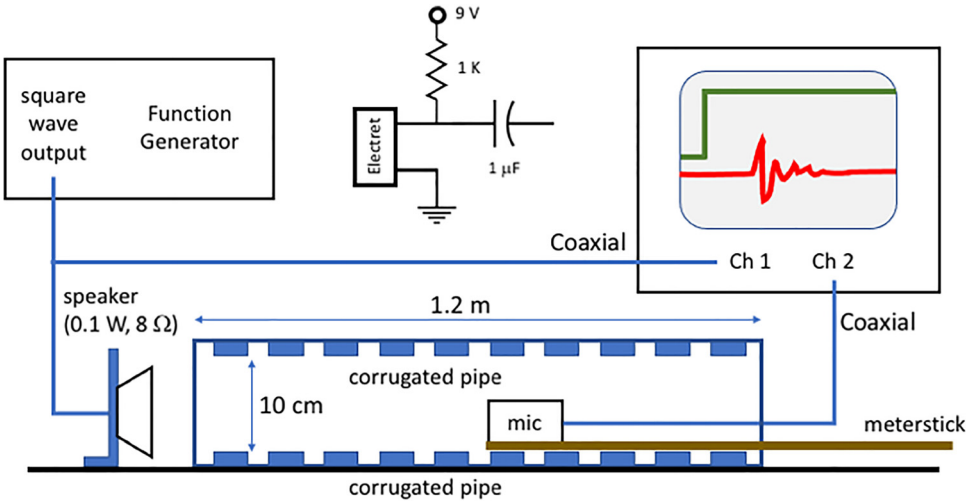


Fig. 1. An outline of the time of flight experiment as discussed in the text with pertinent dimensions provided. Sound pulses from a small speaker travel inside a corrugated pipe arriving at the microphone after a time delay that varies with the position of the microphone. Also shown is the circuit used for constructing a simple microphone from an electret element.

square wave and the time base adjusted to best display both signals together so that the time delay can be most accurately measured. Beginning at some arbitrary separation between the microphone and speaker, the microphone is moved in small, fixed increments (say, 3.0 or 5.0 cm) and the time delay between the rise of the square wave and the initial response of the microphone is recorded. A graph of the time delay against the position of the microphone, like that shown in Fig. 2, can then be produced whose slope equals the inverse of the speed of sound. The error in both the time delay and the position are of the order 1%.

Prior to reorganization of our sophomore lab, students performed a similar measurement for the speed of light, but which required special pulsed-laser equipment, a long folded pathlength and a rather fast oscilloscope. Much of the student effort in this older activity was devoted to aligning the laser beam and dealing with a somewhat jittery signal from

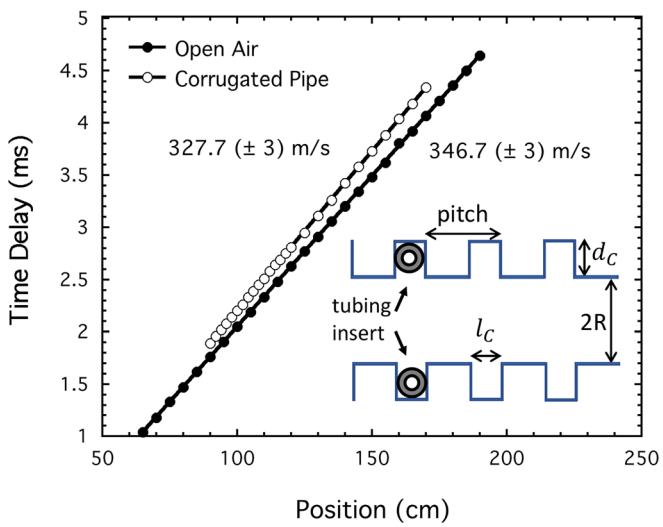


Fig. 2. The measured time delay between pulse generation and reception as a function of arbitrary location of the microphone from the speaker. Measurements in a 10 cm diameter corrugated pipe (open circles) reveal a speed of sound, 328 ± 3 m/s, that is 5% slower than that (347 ± 3 m/s) measured in open air (closed circles). Inset in lower right: The pertinent dimensions of the corrugated pipe discussed in the text.

the detector. Anxiety and frustration were tangible as students struggled to complete the task within the limited time frame provided and this unnecessarily distracted from the intended goal of learning how to use the oscilloscope to accurately measure a known target, the speed of light. By adopting the speed of sound as a target, students also seek to replicate a well-known value (near 343 m/s, depending on temperature and humidity), but the equipment needs become simpler and the experiment can be carried out on a tabletop by multiple groups using generic oscilloscopes.

As the experiment is now performed by multiple groups simultaneously in the same room, the corrugated pipe was originally included as a purely preventative measure to help reduce cross-talk between one group and their neighbors. Sending the sound pulses down the pipe improved the signal intensity and produced a clean, sharp microphone response with little or no jitter. However, as the author himself was surprised to learn during development, and as is evident from Fig. 2, the speed of sound measured in this fashion is only 328 (± 3) m/s, which is about 5% below the anticipated value of 343 m/s (at 20 °C in dry air) and yet distinctly outside the margins of error. Effects due to air humidity are negligible and attempts to account for the discrepancy based on temperature[8] would imply an unreasonably chilly lab temperature of negative 7 °C. Indeed, when the experiment was repeated in the lab *without* the corrugated pipe, the observed 347 (±3) m/s result was seen to be, within error, consistent with the speed of sound in air near 25 °C.

III. PEDAGOGICAL VALUE

At present, we use this lab activity mainly as an opportunity for students to develop skills using an oscilloscope during a single, 3 h lab session. The session starts with some warmup activities, in which students work with the oscilloscope to examine signals from the function generator and from their Arduino circuits practicing how to successfully trigger the signal and adjust the time base appropriately to measure relevant time intervals on the oscilloscope display. After about one hour of warmup, students then proceed to complete the time of flight measurement of the speed of sound within the remaining two hours. Most all complete the

measurements needed and have begun some graphing and error analysis prior to the end of class. The remaining analysis is completed outside of class, and students are asked to draft a simple technical report describing the experiment, the data analysis and error processing, and state their measured speed together with its uncertainty.

Although many students do determine a speed that is smaller than expected, and even recognize that the uncertainty is not the culprit, they rarely believe their result is correct. If asked, many will point to vague sources of potential error that they must have missed including some systematic error hidden in the equipment, such as a mis-calibrated oscilloscope or meterstick. Few are confident enough to conclude that the slower speed is the correct finding. So, although students do gain some skill with using an oscilloscope and some practice doing error analysis and error propagation, they largely fail to appreciate the significance of their result and prefer to assume they have somehow done the experiment incorrectly.

In the spirit of the inquiry-based ideas of Holmes,[1–4] we propose an alternative way in which this speed of sound experiment could be administered so as to promote greater development of critical thinking skills. It begins by extending the activity to a second session, where students are given the opportunity to decide for themselves a procedure for resolving what to them appears to be an incorrect finding caused by some unseen error in their earlier measurement. Following the first lab session, whose format remains as described above, students are asked to elaborate in their technical reports about possible, but *realistic* reasons why their measured speed might be too high or too low. Temperature and humidity are offered as possible things they should consider as well as anything that might be unsecured or misaligned. Students should then propose a plan for how they will use the additional lab session to resolve the errant measurement.

Most students will begin their second session by repeating the measurement, albeit with greater care and diligence, but will be dismayed to find the same speed. Upon conferring with other groups, students will begin to rule out the possibility of mis-calibrated equipment (assuming not every group's equipment is defective too) and begin to accept the validity of the (now collective) result. The class as a whole can stop to speculate about why the speed is slower and brainstorm on some other experiment that might help answer the riddle. Eventually, someone will suggest that the pipe itself might affect the speed and that this could be tested by performing the time of flight measurement again, but without the pipe. When students then observe that the speed in open air is the "correct" value, they finally begin to believe in what they have measured.

Many of us who have taught upper division lab courses in a traditional way with a content-oriented focus will be resistive to devoting an additional lab session to what might appear to be merely repeating the previous week's experiment. However, if one's goal is to deeply influence student thinking about the raw nature of scientific experimentation—the puzzling and uncertain nature—then this additional time is essential and should be viewed as time well spent. As Holmes has repeatedly stressed,[1,2] students need both the autonomy to make their own decisions as well as ample time to reflect on those decisions to succeed in developing scientific prowess.

IV. EXTENSIONS

After students have conceded that the speed of sound in the pipe is slower than in open air, their explorations can be extended into several possible directions. For example, in our Research Methods course, much of the second half of the semester is devoted to project labs, in which students work in small groups to design and construct an apparatus for conducting research on some topic. In one study, a group looked into the effect of changing the pipe diameter on the speed of sound. Their study, however, investigated the speed in *non-corrugated* (i.e., smooth, PVC) pipes with diameters ranging from 10 cm to as small as 2.5 cm (using a miniature electret element) and observed the speed of sound was largely independent of the pipe diameter, remaining near that value observed in open air. Just to confirm their null result, they measured the speed in a length of 7.5 cm diameter corrugated pipe to be slowed to about 325 m/s, similar to, but slightly smaller than that seen in the 10 cm corrugated pipe. To date, the consensus seems to be that the slowing down of the sound pulse in a pipe is due to its corrugated shape.

Theory for why corrugation might decrease the speed of sound draws from a handful of publications[9–11] pertaining to "boomwhackers." These corrugated tubes emit a resonant frequency of sound—a whistling—when air is flowing rapidly through them as they swing about in a circle. Many of these publications reference a treatise by Elliott,[9] in which he outlines a derivation attributed to "Professor A. Cummings" but for which no citation could be located. The model by Cummings is mainly a treatment of one-dimensional waves traveling along the axis of a pipe that has a spring-like admittance along the corrugated surface, and a complete derivation has been reconstructed in the Appendix. Each cavity responds to pressure fluctuations in the fashion of a Helmholtz oscillator, whose resonant frequency is determined by the surface area, $S_C = 2\pi R l_C$, and volume, $V_C = S_C d_c$, of the cavity (see inset of Fig. 2). The wave equation is modified by this average admittance along the walls of the pipe causing the sound speed to decrease as

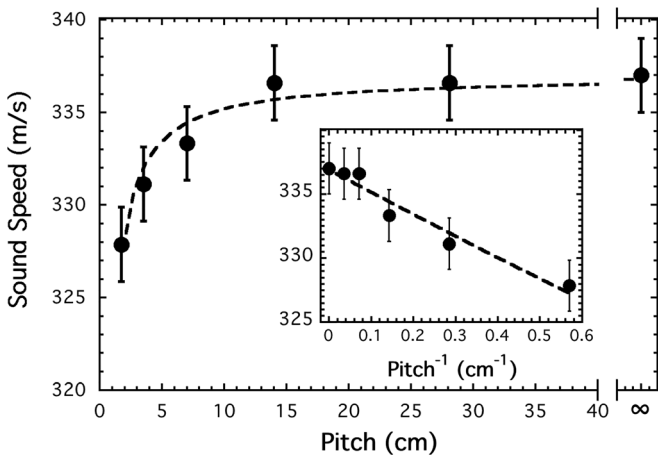


Fig. 3. The speed of sound measured in a 4 in. diameter corrugated pipe for which the pitch (spacing between cavities) was altered using polyvinyl tubing as a filling material. Dashed line is a curve fit using Eq. (1) with $c_o = 337\,\mathrm{m/s}$, and $V_C/A = 0.106\,\mathrm{cm}^2$. Inset shows the same measurements and fit as a function of the inverse of the pitch.

$$c = c_o\left(1+\frac{V_C}{Ap_C}\right)^{-1/2}, \tag{1}$$

where $A = \pi R^2$ is the inner cross-sectional area of the pipe and p_C is the "pitch," the distance between successive cavities, as illustrated in the inset of Fig. 2. This formula suggests a possible project lab investigation of the corrugated pipe ($R = 5.3$ cm, $l_C \approx d_C \approx 0.8$ cm), in which the speed might be measured as the pitch is *systematically* increased by filling in alternating cavities. In one attempt by the author, vinyl tubing (0.8 cm outer diameter) was cut into segments appropriately sized to just expand into, and *mostly* fill, the cavities of the 10 cm diameter corrugated pipe. By adjusting which cavities were filled and which unfilled, an attempt was made to control the pitch while still maintaining both V_C and A of Eq. (1) fixed. The results shown in Fig. 3 suggest some general conformance to the prediction. However, the approach to a limiting value of 337 m/s (instead of 343 m/s) as all the cavities become filled with tubing suggests the tubing did not fully "smooth" the pipe to the extent that is needed to attain the open air value seen in smooth (PVC) pipes.

V. SUMMARY

The speed of sound in corrugated pipes is roughly 5% slower than that measured in open air and provides a unique opportunity for teaching students in the upper division undergraduate labs tangible experimental skills in data analysis and thinking like a physicist. Possible extended explorations have been outlined which are congruent with recent AAPT recommendations that undergraduate labs should focus on the development of skills and habits of a scientist as opposed to merely reinforcing topical content.

APPENDIX: DERIVATION OF EQ. (1) BASED ON CUMMINGS' ACOUSTIC MODEL

Pressure fluctuations, p', are governed by a three-dimensional wave equation

$$\frac{1}{c_o^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0, \tag{A1}$$

along with the second law

$$\rho_o \frac{\partial \vec{u'}}{\partial t} = -\nabla p', \tag{A2}$$

where c_o is the speed of sound, ρ_o is the density of the fluid, and $\vec{u'} = u'\hat{z} + v'\hat{r}$ is the velocity (in cylindrical coordinates). In Cummings' approach, these equations are considered for an interior volume of the tube (see dotted lines in Fig. 4) for which the corrugations can be treated as a smaller surface effect.

Expanding both equations into cylindrical coordinates

$$\frac{1}{c_o^2}\frac{\partial^2 p'}{\partial t^2} - \left\{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p'}{\partial r}\right) + \frac{\partial^2 p'}{\partial z^2}\right\} = 0, \tag{A3}$$

$$\rho_o \frac{\partial u'}{\partial t} = -\frac{\partial p'}{\partial z}, \tag{A4}$$

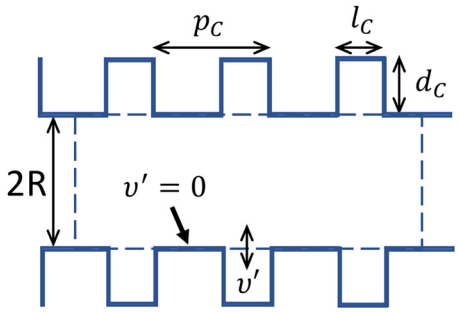


Fig. 4. Illustration of the Cummings model.

$$\rho_o \frac{\partial v'}{\partial t} = -\frac{\partial p'}{\partial r}, \tag{A5}$$

we can then substitute Eqs. (A4) and (A5) into Eq. (A3) to arrive at a nearly 1d form

$$\frac{1}{c_o^2}\frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial z^2} = -\rho_o \frac{\partial}{\partial t}\left\{\frac{v'}{r} + \frac{\partial v'}{\partial r}\right\} = -\rho_o \frac{\partial}{\partial t}\{\nabla \cdot \vec{v'}\}. \tag{A6}$$

The quasi-one dimensional form is obtained by averaging the bracketed quantity on the RHS over the control volume. Upon using the divergence theorem, one sees that

$$\langle \nabla \cdot \vec{v'} \rangle = \frac{1}{AL}\int_{volume} \nabla \cdot \vec{v'} d^3 r = \frac{P}{A}\frac{\int_{wall} v' dz}{L} = \frac{P}{A}\langle v'_w \rangle, \tag{A7}$$

and Eq. (A6) becomes

$$\frac{1}{c_o^2}\frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial z^2} = -\rho_o \frac{P}{A}\frac{\partial}{\partial t}\langle v'_w \rangle. \tag{A8}$$

For one-dimensional wave propagating along the axis of the tube,

$$(p', v', \xi') = (\overline{p}, \overline{v}, \overline{\xi}) \exp[i(kz - \omega t)], \tag{A9}$$

where $v' = \partial \xi'/\partial t$, and substitution into Eq. (A8) develops a dispersion relationship

$$-\frac{\omega^2}{c_o^2} + k^2 = +i\omega\rho_o \frac{P}{A}\frac{\langle \overline{v}_w \rangle}{\overline{p}},$$

or

$$k'^2 \equiv \frac{c_o^2}{c^2} = 1 + i\frac{c_o^2 \rho_o}{\omega}\frac{P}{A}\frac{\langle \overline{v}_w \rangle}{\overline{p}}, \tag{A10}$$

where the quantity $\langle \overline{v}_w \rangle / \overline{p}$ is referred to as the average admittance of the wall. Over regions without a cavity, the wall (dotted line in Fig. 4) is rigid and the admittance is zero. However, over regions (of length l_C) where a cavity is located, the response mimics that of a Helmholtz oscillator. A small mass $m = \rho_o S_C x$ near the interface oscillates in response to a driving force $F' = p' S_C$ where $S_C = P l_C$ is the surface area of the cavity and $V_C = S_C d_C$ is the volume of the cavity. The displacements are governed by

$$m\frac{\partial^2\xi'}{\partial t^2}+m\omega_o^2\xi'=p'S_C, \tag{A11}$$

where resonant frequency is given by

$$\omega_o=c_o\sqrt{\frac{S_C}{V_Cx}}. \tag{A12}$$

The amplitude of displacements, in the limit of low frequencies, is given as

$$\overline{\xi}=\frac{\overline{p}S_C}{m\left(\omega_o^2-\omega^2\right)}\approx\frac{\overline{p}S_C}{m\omega_o^2}, \tag{A13}$$

so the admittance at a cavity is

$$\frac{\overline{v}}{\overline{p}}=\frac{-i\omega\overline{\xi}}{\overline{p}}\approx\frac{-i\omega V_Cx}{\rho_oS_Cxc_o^2}=\frac{-i\omega}{\rho_oc_o^2}\frac{V_C}{S_C}. \tag{A14}$$

Averaging along the wall,

$$\frac{\langle\overline{v}_w\rangle}{\overline{p}}=\frac{-i\omega}{\rho_oc_o^2}\frac{V_C}{S_C}\frac{l_C}{p_C}, \tag{A15}$$

where p_C is the pitch (see Fig. 4). Then,

$$k'^2=\frac{c_o^2}{c^2}=1+\frac{c_o^2\rho_o}{\omega}\frac{P}{A}\frac{\omega}{\rho_oc_o^2}\frac{V_C}{S_C}\frac{l_C}{p_C}=1+\frac{V_C}{Ap_C}, \tag{A16}$$

or

$$c=\frac{c_o}{\sqrt{1+\dfrac{V_C}{Ap_C}}}. \tag{A17}$$

[1]N. G. Holmes and C. E. Wieman, "Introductory physics labs: We can do better," Phys. Today **71**(1), 38–45 (2018).

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[6]American Association of Physics Teachers, *AAPT Recommendations for the Undergraduate Physics Laboratory Curriculum* (2014). <https://www.aapt.org/Resources/upload/LabGuidlinesDocument_EBendorsed_nov10.pdf>

[7]The corrugated pipe is made of high density polyethylene (HDPE) and can be found at most any home improvement store for less than 4 USD per meter.

[8]Wikipedia, <https://en.wikipedia.org/wiki/Speed_of_sound>. The speed of sound varies linearly over temperature ranges from −80 to +50 °C and can be approximated as $331.4+0.606T$, where T is in Celsius.

[9]J. W. Elliot, "Corrugated pipe flow," in *Lecture Notes on the Mathematics of Acoustics*, edited by M. C. M. Wright (Imperial College Press, London, 2005).

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